EMERGENT

FLANDERS TECHNOLOGY & INNOVATION

DATA 4 GOOD CHALLENGE

October 28th



Location: OH Leuven

McKinsey & Company

aws

KBC

RISK CONCILE

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1. The Challenge

Enabling On-Premises Document Retrieval in Crisis-Affected and Vulnerable Areas

Problem statement

Doctors Without Borders is committed to providing emergency medical aid to people in crisis. To this end, it runs humanitarian projects in more than 70 countries, often performing its work under difficult circumstances in crisis-affected and vulnerable areas. In such circumstances, reliable and fast on-premises access to relevant information is crucial.

This challenge tasks participants with designing a document retrieval tool to support doctors operating in crisis-affected and vulnerable areas in reliably and quickly identifying relevant documents. Ideally, this tool can be fully run on-premises (that is, on location), successfully identifies relevant documents, returns a minimal number of false positives, is cost-efficient, and does not create dependency on a specific vendor.

Challenge goal

Doctors Without Borders has provided a collection of free-text documents from practice, with topics ranging from medicine to administrative affairs, and with locations spread out over different continents. The goal is to design a document retrieval tool that identifies relevant documents when provided with a textual query. The document retrieval step may, but does not need to be, supported by a document preprocessing step.

An ideal tool would be:

- Effective: It would reliably and quickly identify relevant documents.
- Cost-effective: Its cost-per-query, implementation cost and maintenance cost would be limited.
- Maintainable: Its design and dependencies would be amenable to updates and changing requirements from the field.
- Usable on-premises: It would be usable in locations with poor internet connection, limited computer storage capacity, and limited computing power. Note that document preprocessing does not need to be performed on-premises.

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2. About our Challenge Partner

Meet this edition's challenge partner

Doctors Without Borders

(FR: Médecins Sans Frontières; NL: Artsen Zonder Grenzen)

Doctors Without Borders (DWB) is an international humanitarian medical organization that provides medical care to those affected by war, epidemics, natural disasters, and extreme poverty. Its work is based on medical need, regardless of nationality, religion, political views, or background. Doctors and nurses work alongside psychologists, logisticians, data scientists, and other specialists.

DWB was founded in France in 1971 by doctors and journalists who had witnessed the devastating Biafran War and famine in Nigeria. They wanted to create an organization that could respond quickly to humanitarian crises and speak openly about what its teams encountered. From the beginning, DWB has been guided by three principles: impartiality, neutrality, and independence. These mean providing assistance according to need, avoiding political sides, and remaining free from outside interests.

In the past decades, DWB has grown into a worldwide humanitarian movement with over 65,000 people. Its teams have responded to conflicts, refugee crises, epidemics, natural disasters, and disease outbreaks, while also working on issues such as malnutrition, maternal healthcare, vaccination, and mental health. In 1999, DWB received the Nobel Peace Prize for its humanitarian work. Today, its teams operate in more than 70 countries, often in crisis-affected and vulnerable areas. Its core principles remain providing medical care to people who need it most while maintaining the independence to act and speak freely.

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3. The Data

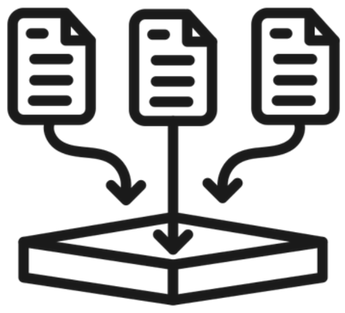
Dig in & gain some insights!

This year's challenge uses a collection of free-text documents supplied by Doctors Without Borders. Two types of data are provided, namely original files (*.html* and *.pdf*) and Markdown files (*.md*).

The Markdown files contain all text from the HTML and PDF files in a more easily-digestible format, but not all images and figures in the PDF version may have been parsed well. The Markdown files are hence easier to work with, but solutions starting from the HTML and PDF files may be more effective.

4. Deliverables

Challenge deliverables at a glance



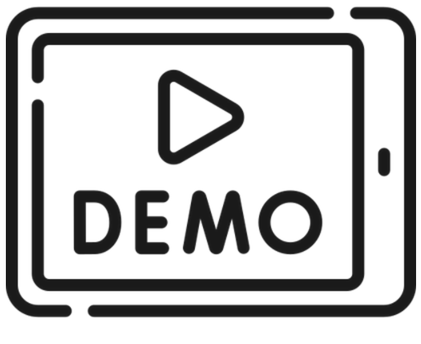
Document retrieval pipeline

A modular, well-documented codebase that implements the complete document retrieval pipeline, potentially including a document preprocessing step.

Presentation

Slides, visualization or demo to present your solution, its technical core and key design choices in a pitch of no more than 5 minutes. Optionally, you may also add a demo snapshot or link if available.





Optional:

Notebook or lightweight app

An easily-accessible and user-friendly implementation of your document retrieval pipeline in the form of a notebook or lightweight app.

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5. Timeline

Time	Activity
8h - 9h	Breakfast
9h - 10h	Welcome presentation Emergent / D4GC Intro FTI & partners presentation
10h - 12h	Partner workshops
12h - 13h	Lunch
13h - 14h	Challenge presentation
14h - 18h	Challenge
18h - 19h	Dinner
18h30 - 20h	Pitching
20h30 - 22h30	Award ceremony & reception

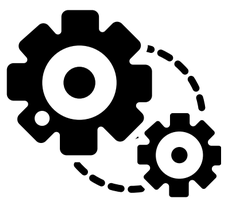
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6. Scoring

How will you be evaluated?

All solutions will be evaluated against the following five criteria by the jury:



Effectiveness

The solution is able to reliably and quickly identify relevant documents while limiting false positives



Creativity

Show what is original and useful about the approach. Explain what new insights it creates and how it can adapt to changing requirements or document collections.



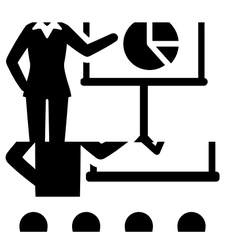
Implementability

The solution is easy to implement and maintain in practice. It can be used in locations with limited computer storage and poor internet connection, and it is not overly reliant on software or other solutions owned by a single vendor.



Cost-effectiveness

The solution's cost per query, implementation costs and maintenance costs are reasonable given its level of effectiveness.



Presentation

Deliver a clear, structured pitch of no more than 5 minutes. Explain the retrieval procedure, optional preprocessing, key insights and design choices, supported by a demo snapshot or link when available.

7. FAQ

Frequently asked questions

Should our solution work with the full document collection?

Yes. The retrieval pipeline should search the supplied collection. Preprocessing may be used to improve speed and reliability.

Do we have to use a particular software or product?

No. Choose the tools that best support your approach, provided the solution can run on premises, remains maintainable and avoids dependency on a specific vendor.

Can we use external data or models?

Yes, if they are clearly sourced and compatible with the on-premises, cost and vendor-independence requirements.

May document preprocessing run off-premises?

Yes. Only document retrieval needs to be usable on-premises. Preprocessing may be performed elsewhere.

Do we need to prepare a presentation?

Yes. Prepare a concise, presentation-style pitch of no more than 5 minutes and submit it with your codebase and, optionally, a demonstration notebook or lightweight app.

Does everyone from the team need to present to the jury?

No, you may decide among yourselves, but at least one person should present.

How much time do we have to present?

Your presentation-style pitch may last no more than 5 minutes.